HK IMO Training 2022-2023
Problem Set 4 Suggested Solutions

Problem 13 (Dutch MO 2016 Problem 5). We colour all positive integers such that the following three conditions hold:

- all the odd positive integers are red;
- n has the same colour as $4n$ for every positive integer n;
- n has the same colour as $n+2$ or $n+4$ for every positive integer n.

Prove that all positive integers are red.

Solution. Suppose on the contrary that not all positive integers are red. We pick the smallest one which is not red. It must be even, say $2k$ is blue for some $k \geqslant 1$. Then $8k = 4(2k)$ is blue. As $8k+4 = 4(2k+1)$ is red, $8k+2$ must be blue, or otherwise $8k$ does not have the same colour as both $8k+2$ and $8k+4$. Now, as $8k = (8k-2)+2$ and $8k+2 = (8k-2)+4$ are blue, we know that $8k-2$ is blue. Similarly, $8k-4$ is blue since both $8k-2$ and $8k$ are blue. But then $8k-4 = 4(2k-1)$ should be red, contradiction. So all positive integers are red. $\square$

Problem 14 (Rioplatense MO 2009 Problem 4). Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy) = \max\{f(x+y), f(x)f(y)\} \tag{1}$$

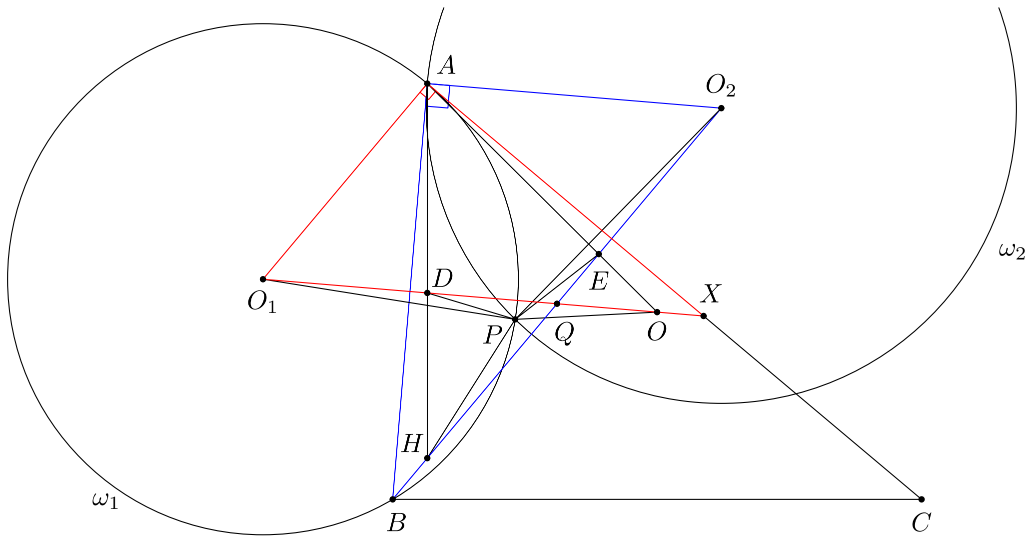
for all real numbers x and y.

Solution. Putting $y = 0$ in (1), we obtain $f(0) = \max\{f(x), f(x)f(0)\}$. This implies

$$f(0) \geqslant f(x) \tag{2}$$

for all $x \in \mathbb{R}$. Putting $y = -x$ in (1), we obtain $f(-x^2) = \max\{f(0), f(x)f(-x)\}$. This implies $f(-x^2) \geqslant f(0)$ for all $x \in \mathbb{R}$. Using (2), we see that $f(0) \geqslant f(-x^2) \geqslant f(0)$, and thus $f(-x^2) = f(0)$ for all $x \in \mathbb{R}$. Next, for every $x < 0$, by putting $y = x$ in (1), we have $f(x^2) = \max\{f(2x), f(x)^2\}$. This implies $f(x^2) \geqslant f(2x) = f(0)$ since $2x < 0$. Using (2), we see that $f(0) \geqslant f(x^2) \geqslant f(0)$, and thus $f(x^2) = f(0)$ for all $x \in \mathbb{R}$. It follows that f is a constant function. Let $f(x) = c$ for all $x \in \mathbb{R}$. Then (1) is the same as $c = \max\{c, c^2\}$. This holds if and only if $c \geqslant c^2$, i.e. $0 \leqslant c \leqslant 1$. Therefore, the solutions are the constant functions $f(x) = c$ with $0 \leqslant c \leqslant 1$. $\square$

Problem 15 (Czech-Polish-Slovak Match 2015 Problem 5). Let $\triangle ABC$ be a non-equilateral triangle with orthocentre H and circumcentre O. Let ω_1 be the circle passing through B and tangent to AC at A. Let ω_2 be the circle with centre lying on the line BH and tangent to AB at A. The circles ω_1 and ω_2 meet again at P. Prove that $\angle HPO = 180° - \angle BAC$.



Solution. Let BH meet OO_1 and AO at Q and E respectively. Let OO_1 meet AH and AC at D and X respectively. Note that AO_1QO_2 is a parallelogram since both AO_1 and O_2Q are perpendicular to AC, while both AO_2 and O_1Q are perpendicular to AB. Thus, we have $\angle AO_2B = \angle AO_1X$. Together with $\angle BAO_2 = 90° = \angle XAO_1$, we have $\triangle ABO_2 \sim \triangle AXO_1$. Also, note that $\angle BAH = \angle XAO$ and $\angle BAE = \angle XAD$ due to the isogonal lines AH and AO. Therefore, we have $ABHEO_2 \sim AXODO_1$. Next, we have $\dfrac{PO_1}{PO_2} = \dfrac{AO_1}{AO_2} = \dfrac{O_1D}{O_2E}$ (from the above similarity). Also, we have

$$\begin{aligned}\measuredangle PO_1D - \measuredangle PO_2E &= (\measuredangle PO_1A - \measuredangle OO_1A) - (\measuredangle AO_2E - \measuredangle AO_2P)\\ &= 2\measuredangle O_2O_1A - \measuredangle BAC - \measuredangle BAC + 2\measuredangle AO_2O_1\\ &= 2(\measuredangle O_2O_1A + \measuredangle AO_2O_1) - 2\measuredangle BAC\\ &= 2\measuredangle O_2AO_1 - 2\measuredangle BAC\\ &= 0.\end{aligned}$$

This shows $\triangle PO_1D$ and $\triangle PO_2E$ are directly similar, and so P is the centre of spiral similarity mapping O_1DOX to O_2EHB. This yields $\measuredangle POO_1 = \measuredangle PHO_2$, and hence P, H, O, Q are concyclic. Finally, this implies

$$\angle HPO = \angle HQO = \angle O_2QO_1 = \angle O_1AO_2 = 180° - \angle BAC.$$

□

Problem 16 (International Zhautykov Olympiad 2016 Problem 6)**.** Let α and β be two irrational numbers. Let A be the set of all positive integers n such that $\left|\alpha - \dfrac{m}{n}\right| < \dfrac{1}{10n}$ for some integer m. Let B be the set of all positive integers n such that $\left|\beta - \dfrac{m}{n}\right| < \dfrac{1}{10n}$ for some integer m. If $A = B$, show that $\alpha + \beta$ or $\alpha - \beta$ is an integer.

Solution. Note that $\left|\alpha - \dfrac{m}{n}\right| < \dfrac{1}{10n}$ is the same as $|n\alpha - m| < 0.1$. Thus, $n \in A$ iff $\{n\alpha\} \in (0, 0.1)$ or $\{n\alpha\} \in (0.9, 1)$. We partition A into two sets A_1, A_2 such that $n \in A_1$ iff $\{n\alpha\} \in (0, 0.1)$ and $n \in A_2$ iff $\{n\alpha\} \in (0.9, 1)$. We partition B into B_1, B_2 analogously. Note that A_1, A_2 are nonempty (similar for B_1, B_2) since $\{n\alpha\}$ is dense in $(0, 1)$ for irrational α.

Consider any $m, n \in A_1$. We claim that both m, n belong to B_1 or both of them belong to B_2. Suppose on the contrary that $m \in B_1$ and $n \in B_2$.

We show that for every $k \in \mathbb{N}$, there exist integers $i, j \geqslant 0$ such that $i + j = k$ and $im + jn \in B$. Indeed, for the base case $k = 1$, we can simply take $i = 1$ and $j = 0$. Suppose we have found $i, j \geqslant 0$ such that $i + j = k$ and $im + jn \in B$. If $im + jn \in B_1$, then

$$\{(im + (j + 1)n)\beta\} = \{(im + jn)\beta + n\beta\} \in (0, 0.1) \cup (0.9, 1)$$

since $\{(im + jn)\beta\} \in (0, 0.1)$ and $\{n\beta\} \in (0.9, 1)$. This implies $im + (j + 1)n \in B$, where $i + (j + 1) = k + 1$. If $im + jn \in B_2$, then

$$\{((i + 1)m + jn)\beta\} = \{(im + jn)\beta + m\beta\} \in (0, 0.1) \cup (0.9, 1)$$

since $\{(im + jn)\beta\} \in (0.9, 1)$ and $\{m\beta\} \in (0, 0.1)$. This implies $(i + 1)m + jn \in B$, where $(i + 1) + j = k + 1$. The result holds by induction.

As every such $im + jn \in B$, we must have $im + jn \in A$. Each time when k is increased by 1, the fractional part of $(im + jn)\alpha$ is increased by $\{m\alpha\}$ or $\{n\alpha\}$, both of which are in $(0, 0.1)$. Therefore, $im + jn \notin A_2$, and hence $im + jn \in A_1$. But this is also impossible as $\{(im + jn)\alpha\} = i\{m\alpha\} + j\{n\alpha\} \in (0, 0.1)$, but one of i and j is unbounded. This is a contradiction.

Similarly, for any $m, n \in A_2$, both m, n belong to the same of B_1 and B_2. Note that we may replace β by $1 - \beta$ without affecting the result. Therefore, we may now assume $A_1 = B_1$ and $A_2 = B_2$. Next, define $A_{1,k}$ to be the set of all n satisfying $\{n\alpha\} \in \left(\dfrac{1}{10(k+1)}, \dfrac{1}{10k}\right)$, and define $B_{1,k}$ analogously. Then $A_{1,1}, A_{1,2}, \ldots$ is a partition of A_1. Again, each $A_{1,k}$ and $B_{1,k}$ is nonempty. Note that we also have $A_{1,k} = B_{1,k}$. Otherwise, up to symmetry, we can find some n, k, ℓ such that $n \in A_{1,k}$, $n \in B_{1,\ell}$ and $k < \ell$. But then we easily check that $(k + 1)n \notin A$ while $(k + 1)n \in B$, contradicting $A = B$.

Suppose $\{\alpha\} \neq \{\beta\}$. If $1 \in A_1 = B_1$, define $r = 0.1 - \{\alpha\}$ and $s = 0.1 - \{\beta\}$. WLOG assume $s > r > 0$. Let k be sufficiently large so that $10(s - r)k > 10r + 1$. This implies

$$10r(k + 1) + 1 < 10sk.$$

We pick any $m \in A_{1,k} = B_{1,k}$, and let $d = \lceil 10r(k + 1)\rceil$. We check that $1 + md \notin A_1$ but $1 + md \in B_1$. Indeed, we have

$$\{(1 + md)\alpha\} = \{\alpha\} + d\{m\alpha\} > (0.1 - r) + \frac{d}{10(k + 1)} \geqslant (0.1 - r) + r = 0.1$$

and

$$\{(1+md)\beta\} = \{\beta\} + d\{m\beta\} < (0.1-s) + \frac{d}{10k} < (0.1-s) + s = 0.1.$$

This yields the contradiction that $A_1 \neq B_1$. Similarly, we can obtain a contradiction if $1 \in A_2 = B_2$, and if $1 \notin A = B$ (since we can find some large k, some $m \in A_{1,k}$ and some suitable d such that $1 + md$ belongs to exactly one of A and B). Thus, we must have $\{\alpha\} = \{\beta\}$, and hence $\alpha - \beta \in \mathbb{Z}$. As we may have replaced β by $1 - \beta$, another possibility is that $\alpha + \beta \in \mathbb{Z}$. $\square$